

预处理的艺术

前缀和、差分与双指针 · 7题 · C++ & Python 双语对照

NQ108 AcWing 795 AcWing 795

AcWing 795

输入 见原题**输出** 见原题**范围** 见原题输入
见原题输出
见原题**思路** 见AcWing原题描述**技巧** 参考模板库 CONTEXT.md**C++**

```
#include <iostream>

using namespace std;
const int N = 1000007;
int n, m;

int a[N], s[N];

int main()
{
    scanf("%d%d", &n, &m);
    for (int i = 1; i <= n; i++)
        scanf("%d", &a[i]);
    s[0] = 0; //刻意初始化s[0]为0，此句可以不写，写也无妨
    for (int i = 1; i <= n; i++)
        s[i] = s[i - 1] + a[i]; //打表法
    //m次询问
    while (m--)
    {
        int l, r;
        scanf("%d%d", &l, &r);
        printf("%d\n", s[r] - s[l - 1]); //求区间和，直接从s中读取数值
    }

    return 0;
}
```

Python

AcWing 795

NQ109 AcWing 796 AcWing 796

AcWing 796

输入 见原题**输出** 见原题**范围** 见原题

输入

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输出

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思路 见AcWing原题描述**技巧** 参考模板库 CONTEXT.md

C++

```

#include <iostream>
using namespace std;
const int N = 1007;
int n, m, q;
int a[N][N], s[N][N]; //子矩阵

int main()
{
    //读入矩阵
    scanf("%d%d%d", &n, &m, &q);
    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= m; j++)
            scanf("%d", &a[i][j]);

    s[0][0] = 0; //刻意初始化s[0]为0，此句可以不
    //写，写也无妨
    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= m; j++)
            s[i][j] = s[i - 1][j] + s[i][j - 1]
            - s[i - 1][j - 1] + a[i][j]; //打表法
    //q次询问
    while (q--)
    {
        int x1, x2, y1, y2;
        scanf("%d%d%d%d", &x1, &y1, &x2, &y2);
        //求子矩阵和，直接从s中读取数值
        printf("%d\n", s[x2][y2] - s[x2][y1 - 1]
        - s[x1 - 1][y2] + s[x1 - 1][y1 - 1]);
    }
    return 0;
}

```

Python

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AcWing 797

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思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
#include <iostream>
using namespace std;
const int N = 1000007;
int n, m;
int a[N], b[N];

//差分的插入公式
void insert(int l, int r, int c)
{
    b[l] += c;
    b[r + 1] -= c;
}

int main()
{
    cin >> n >> m;
    //读入数组
    for (int i = 1; i <= n; i++)
        cin >> a[i];
    //构造差分数组b
    for (int i = 1; i <= n; i++)
        insert(i, i, a[i]);
    //输出查询结果
    while (m--)
    {
        int l, r, c;
        cin >> l >> r >> c;
        insert(l, r, c);
    }
    //重新复原a数组(前缀和)
    for (int i = 1; i <= n; i++)
        a[i] = a[i - 1] + b[i];
    //输出a
    for (int i = 1; i <= n; i++)
        cout << a[i] << " ";
    cout << endl;
    return 0;
}
```

Python

AcWing 797

AcWing 798

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思路 见AcWing原题描述**技巧** 参考模板库 CONTEXT.md

C++

```

#include <iostream>
using namespace std;
const int N = 1007;
int n, m, q;
int a[N][N], b[N][N]; //子矩阵
//二维差分核心操作
void insert(int x1, int y1, int x2, int y2, int c)
{
    b[x1][y1] += c;
    b[x1][y2 + 1] -= c;
    b[x2 + 1][y1] -= c;
    b[x2 + 1][y2 + 1] += c;
}
int main()
{
    //读入矩阵
    scanf("%d%d%d", &n, &m, &q);
    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= m; j++)
            scanf("%d", &a[i][j]);

    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= m; j++)
            insert(i, j, i, j, a[i][j]); //构造
    二维差分矩阵

    //q次询问
    while (q--)
    {
        int x1, x2, y1, y2, c;
        scanf("%d%d%d%d%d", &x1, &y1, &x2, &y2,
&c);
        //差分核心操作
        insert(x1, y1, x2, y2, c);
    }
    //重建a[][],即求二维前缀和
    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= m; j++)
            a[i][j] = a[i - 1][j] + a[i][j - 1]
- a[i - 1][j - 1] + b[i][j];
    //输出a[][]
    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= m; j++)
            cout << a[i][j] << " ";
        cout << endl;
    }
    return 0;
}

```

Python

AcWing 798

NQ112 AcWing 799

AcWing 799

AcWing 799

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思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
#include <iostream>
using namespace std;
const int N=100007;
int num[N],visited[N];

int main(){
    int n;
    cin>>n;
    for (int i=0; i<n; i++ ) cin>>num[i]; //读入
    n个数
    //双指针扫描
    int res=0; //初始化为最小值
    for (int i=0, j=0; i<n; i++){
        visited[num[i]]++;
        while(j<=i && visited[num[i]]>1){
            visited[num[j]]--;
            j++; //去掉重复数
        }
        res= max(res,i-j+1); //j--i 之间的数字个数
    }
    cout<<res<<endl;
    return 0;
}
```

Python

AcWing 799

NQ113 AcWing 800

AcWing 800

AcWing 800

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思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```

#include <iostream>
#include <algorithm>
using namespace std;
const int N=100007;
int n,m,x;
int numA[N],numB[N];

int main(){
    //读入n,m,x其中x为目标和
    scanf("%d%d%d",&n,&m,&x);
    for (int i=0; i<n; i++) scanf("%d",&numA[i]);
    for (int j=0; j<m; j++) scanf("%d",&numB[j]);
    //双指针查找和

    for (int i=0, j= m-1; i<n; i++){
        //check(i,j)
        while (j>=0 && numA[i]+numB[j]>x) j--;
        if (numA[i]+numB[j]==x)//判断是否和相等
        {
            printf("%d %d\n",i,j);
            break;
        }
    }
    return 0;
}

```

Python

AcWing 800

NQ114 AcWing 2816 AcWing 2816

AcWing 2816

输入

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思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
#include <iostream>
#include <cstdio>
using namespace std;
const int N = 1e5+7; // 我的风格
int a[N], b[N];

int main()
{
    int n, m;
    scanf("%d%d", &n, &m);
    for (int i = 0; i < n; i++) scanf("%d", &a[i]);
    for (int i = 0; i < m; i++) scanf("%d", &b[i]);

    int p = 0; // 指向a数组的指针
    for (int i = 0; i < m; i++)
    {
        if (p < n && a[p] == b[i]) p++;
    }
    // 完全匹配
    if (p == n) puts("Yes");
    else puts("No");

    return 0;
}
```

Python

AcWing 2816